

Multi-threading:-

```
import java.util.LinkedList;
import java.util.Queue;
import java.util.Scanner;
```

```
Class RegistrarParking {
    private static int count=0;
    private final int carId;
```

```
    public RegistrarParking() {
        this.carId = ++count;
    }
```

```
    public int getCarId() {
        return carId;
    }
}
```

```
class ParkingPool {
```

```
    private private final Queue < RegistrarParking > queue = new
        LinkedList < > ();
```

```
    public synchronized void addCar (RegistrarParking car) {
        queue.add (car);
        notifyAll ();
    }
```

```
    public synchronized RegistrarParking getCar ()
        throws InterruptedException {
        while (queue.isEmpty ()) {
            wait ();
        }
        return queue.poll ();
    }
}
```

```

class ParkingAgent extends Thread {
    private final String agentName;
    private final ParkingPool parkingPool;

    public ParkingAgent(String name, ParkingPool pool) {
        this.agentName = name;
        this.parkingPool = pool;
    }

    @Override
    public void run() {
        while (true) {
            try {
                Register a parking car = parkingPool.getCar();
                System.out.println(agentName + " parked car # "
                    + car.getCarId());
                Thread.sleep(1000);
            }
            catch (InterruptedException e) {
                System.out.println(agentName + " interrupted.");
                break;
            }
        }
    }
}

```

```

public class MainClass {
    public static void main(String[] args) {
        ParkingPool pool = new ParkingPool();
        new ParkingAgent("Valet -1", pool).start();
        new ParkingAgent("Valet -2", pool).start();
        new ParkingAgent("Valet -3", pool).start();
        Scanner sc = new Scanner(System.in);
        int carCount = 0;
    }
}

```

```

while (carCount < 10) {
    System.out.println("Press Enter to register a car  
for parking....");
    sc.nextInt sc.nextLine();

    RegistrarParking car = new RegistrarParking();
    pool.addCar(car);
    System.out.println("Car #" + car.getCarId() +  
"registered for parking.");

    carCount++;
}
sc.close();
System.out.println("All cars registered.  
Parking in progress....");
}
}

```